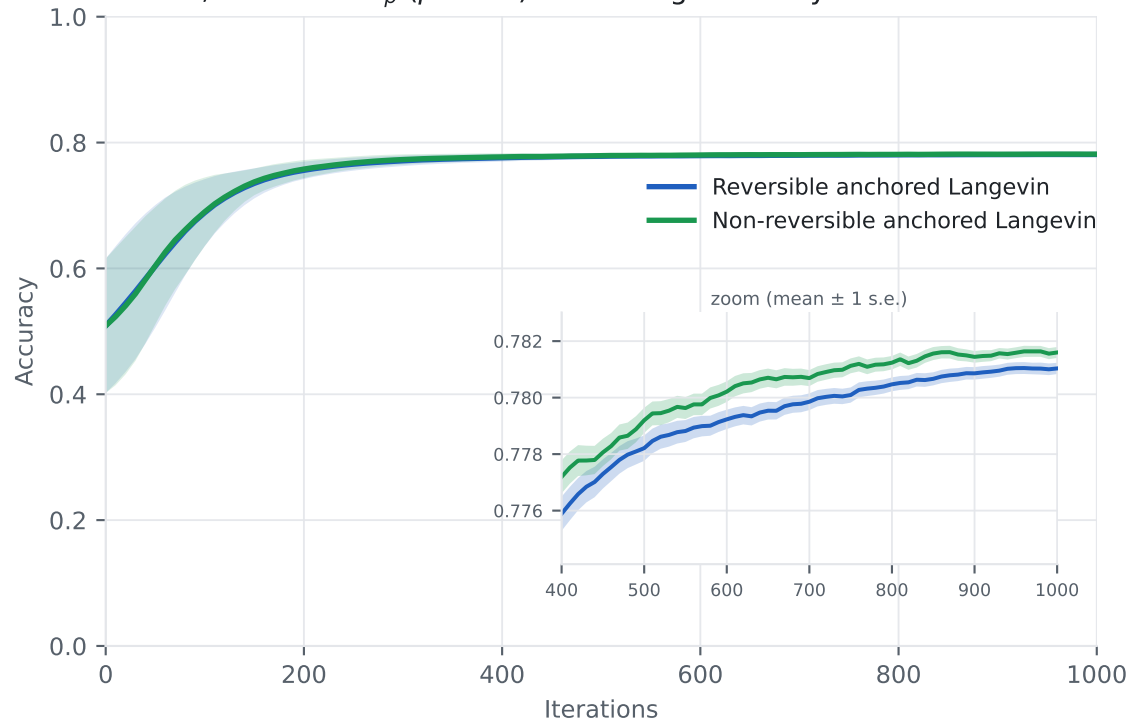
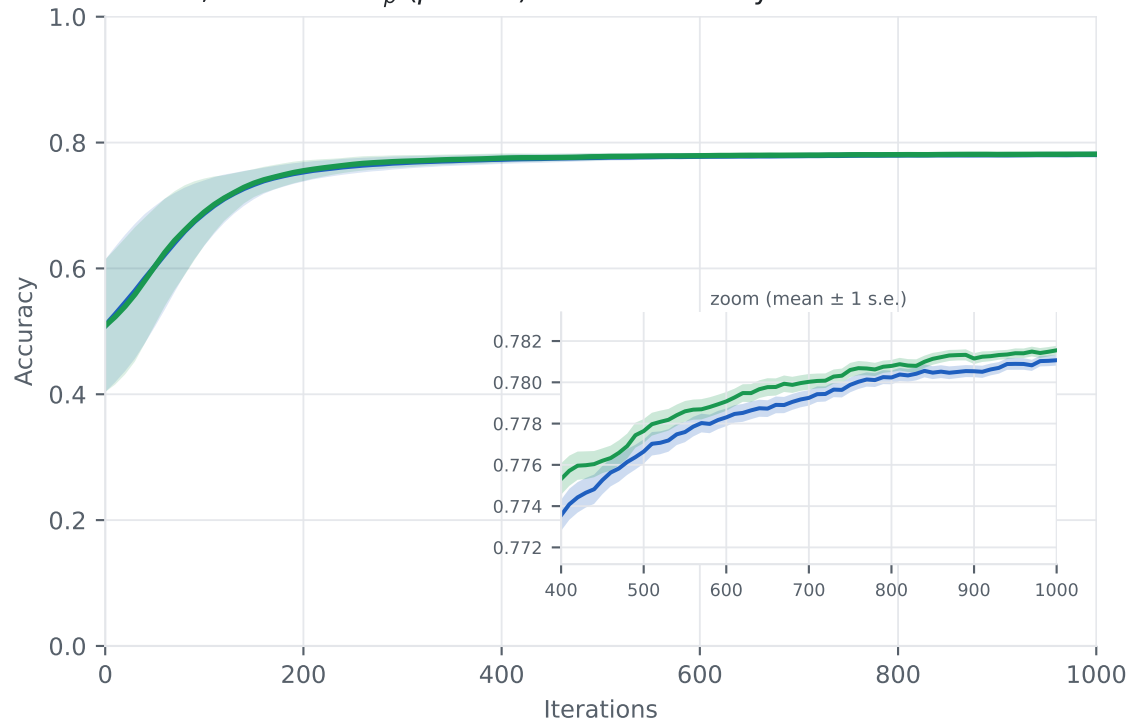


MAGIC, smoothed ℓ_p ($p = 2.4$) — training accuracy



MAGIC, smoothed ℓ_p ($p = 2.4$) — test accuracy



MAGIC -- smoothed ℓ_p , $g(w) = \sum_i (w_i^2 + \text{eps}^2)^{(p/2)} \leq \text{Lambda}$, $p = 2.4$, $\text{eps} = 0.2$, $\text{Lambda} = 4$. $d = 10$ (intercept w_0 + nine features), no mini-batching: the EXACT gradient is used at every step. $U = \sum \text{softplus} - y \cdot x \cdot w + w_0^2 / (2 \sigma^2) + \text{lambda} \sum_{j \geq 1} |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \delta^2}$; $a = \exp(U - U_0)$. $\sigma = 10$, $\text{lambda} = 152.2 = 0.01$. $n_{\text{train}}, \delta = 0.0005062$, so a in $[0.50, 1]$. Block strengths $s = (5, 5, 5)$ on $w_1..w_9$ only. Step $\text{eta} = 1e-06$. $R = 100$ coupled replicates (same W_0 , same Gaussian stream) on one fixed stratified 80/20 split ($n_{\text{train}} = 15216$, $n_{\text{test}} = 3804$). Paired final-iterate difference, non-reversible minus reversible: training $+0.0006$ ($t = +2.96$), test $+0.0005$ ($t = +1.87$). Bands: mean $\pm$ 1 SD (ddof = 1) across replicates -- repeat-run variability at a fixed split, not a confidence or credible interval.